STUDYLAB HUB · AP & COLLEGE GENERAL CHEMISTRY · UNIT 1 (CHAPTERS 1–2)

Unit 1: Atomic Structure & Properties

Moles & Molar Mass · Mass Spec · Empirical Formulas · Electron Configuration · PES · Periodic Trends

AP & College General Chemistry — Unit 1 (Chapters 1–2): Atomic Structure & Properties

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1. Unit 1 Roadmap (Chapters 1–2)

Topic	Core Formula / Principle	Exam Trap to Watch For
1.1 · Moles & Molar Mass	Moles (n) = mass (g) ÷ Molar Mass (g/mol) = Particles (N) ÷ (6.022×10^{23} mol^{-1})	Confusing **molecules/formula units** (e.g., O_2, $CaCl_2$) with total **atoms or ions** (1 mol $CaCl_2$ dissociates into **3 mol total ions**: 1 mol Ca^{2+} + 2 mol Cl^-)
1.2 · Mass Spectroscopy of Elements	Average Atomic Mass = Σ (fractional abundance × isotopic mass)	Never take a simple unweighted average of isotope masses; weight each isotope peak by its relative abundance
1.3–1.4 · Elemental Composition & Empirical Formula	Mass % → Moles → Divide by smallest mole amount → Whole-number ratio	Rounding too early: a mole ratio of 1 : 1.33 must be multiplied by 3 to give X_3Y_4, never rounded to 1 : 1
1.5 · Atomic Structure & Electron Configuration	Aufbau Principle, Pauli Exclusion, Hund's Rule; Coulomb's Law: $F = k \times (q_1 \times q_2) \div r^2$	**Transition metal cations** lose valence **ns electrons BEFORE (n–1)d electrons** (e.g., Fe^{2+} is [Ar] $3d^6$, *not* [Ar] $4s^2$ $3d^4$)
1.6 · Photoelectron Spectroscopy (PES)	Binding Energy = Photon Energy (hν) – Kinetic Energy (KE); peak height is proportional to number of electrons in subshell	Reading the x-axis backwards: PES binding energy usually **decreases** from left to right (highest binding energy peak = 1s core electrons closest to the nucleus)
1.7–1.8 · Periodic Trends & Valence Electrons	Effective nuclear charge (Z_{eff}) vs. occupied principal electron shells (n)	Stating "going right across a period increases ionization energy" as a reason — **the trend is the observation, not the cause!** Always cite **higher Z_{eff} (more protons with same core shielding)** or **shell distance (n)**

2. Coulomb's Law: The Master Key to Every Unit 1 Question

Every periodic trend and PES question is governed by **Coulomb's Law**:

> **Electrostatic Attraction Energy is proportional to (Nuclear Charge × Electron Charge) ÷ Distance (r)**

1. **Across a Period (Left → Right)**:

- Proton count (Z) increases while inner core electrons remain in the same inner shells ⇒ **Effective nuclear charge (Z_{eff}) increases**.
- Stronger attraction pulls the valence shell closer to the nucleus ⇒ **Atomic radius decreases**, **First Ionization Energy (IE_1) increases**, and **Electronegativity increases**.

2. **Down a Group (Top → Bottom)**:

- Valence electrons occupy a higher principal energy level (n = 2 → 3 → 4...) with additional inner core electron shielding ⇒ distance (r) from the nucleus increases.
- Weaker attraction ⇒ **Atomic radius increases**, **First Ionization Energy (IE_1) decreases**, and **Electronegativity decreases**.

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3. Two Famous First Ionization Energy (IE_1) Exceptions

Anomaly	What Happens	Exact Orbital Explanation for FRQs
Group 2 vs. Group 13 (e.g., Be vs. B, or Mg vs. Al)	IE_1(B) < IE_1(Be) even though Boron has 1 more proton	Boron's outermost electron occupies a **2p subshell** ($1s^2\ 2s^2\ 2p^1$), which is slightly higher in energy and partially shielded by the full $2s^2$ subshell, making it easier to remove than a 2s electron from Beryllium
Group 15 vs. Group 16 (e.g., N vs. O, or P vs. S)	IE_1(O) < IE_1(N) even though Oxygen has 1 more proton	Nitrogen ($2p^3$) has three unpaired electrons in separate 2p orbitals (Hund's rule). Oxygen ($2p^4$) forces a fourth electron to **pair up in one 2p orbital**, creating **electron–electron repulsion** that makes that paired electron easier to remove

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4. Photoelectron Spectroscopy (PES) Rapid Decoder

- **X-Axis (Binding Energy)**: Measures the energy required to eject an electron from a specific subshell.
- **Highest Binding Energy Peak** = **1s core electrons** (closest to the nucleus, smallest distance r, strongest Coulombic attraction).
- **Lowest Binding Energy Peak** = **Valence electrons** (furthest from the nucleus, highest principal level n, shielded by inner core electrons).
- **Y-Axis (Relative Number of Electrons / Peak Height)**:
- Peak height is directly proportional to the number of electrons in that subshell: if a $1s^2$ peak has height 2, then a $2p^6$ peak is **3 times taller** (height 6), and a $3s^1$ peak (Sodium, Na) is **half as tall** (height 1).
- **Comparing Two Elements' PES Spectra**:
- Because Oxygen (8 protons) has a greater nuclear charge than Nitrogen (7 protons), **every corresponding subshell peak (1s, 2s, 2p) in Oxygen appears at a higher binding energy** than in Nitrogen.

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5. Empirical & Molecular Formula + Hydrate Lab Calculations

Standard Empirical Formula Workflow

Given mass percentages (e.g., 40.0% C, 6.71% H, 53.29% O):

1. Assume a 100.0 g sample → 40.0 g C, 6.71 g H, 53.29 g O.
2. Convert grams to moles:

- Moles of C = 40.0 g ÷ 12.01 g/mol = 3.33 mol C
- Moles of H = 6.71 g ÷ 1.008 g/mol = 6.66 mol H
- Moles of O = 53.29 g ÷ 16.00 g/mol = 3.33 mol O

3. Divide all mole values by the smallest (3.33 mol): $C_1H_2O_1$ → Empirical formula = **CH_2O** (Empirical molar mass = 30.03 g/mol).
4. If the experimental molar mass is 180.16 g/mol, multiplier = 180.16 ÷ 30.03 = 6 → Molecular formula = **$C_6H_{12}O_6$**.

Hydrate Heating Lab ($CuSO_4 \cdot x\ H_2O$)

- **Mass lost upon heating** = mass of evaporated water (H_2O).
- **Remaining solid mass in crucible** = mass of anhydrous salt ($CuSO_4$).
- **Common Lab Error Questions**:
- *If the hydrate is not heated long enough*: some water remains trapped in the crystals → calculated mass of H_2O lost is **too low** → calculated mole ratio x is **too low**.
- *If solid splatters out of the crucible during heating*: lost solid is mistakenly counted as evaporated water → calculated mass of H_2O lost is **too high** → calculated mole ratio x is **too high**.

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6. Worked Exam Practice (MCQs + Scored FRQ)

Practice Question 1 (Isoelectronic Ion Radii)

Rank the following isoelectronic species in order of increasing radius: O^{2-}, F^-, Ne, Na^+, Mg^{2+}. Justify using Coulomb's Law.

- **Answer**: **$Mg^{2+} < Na^+ < Ne < F^- < O^{2-}$**.
- **Justification**: All five species have the exact same electron configuration ($1s^2\ 2s^2\ 2p^6$, 10 total electrons) and identical inner-shell shielding. However, nuclear charge increases from O^{2-} (8 protons) to Mg^{2+} (12 protons). By Coulomb's Law, the 12 protons in Mg^{2+} exert the strongest attractive pull on the 10 electrons, drawing the n = 2 valence shell closest to the nucleus.

Practice Question 2 (Successive Ionization Energies)

Element X in Period 3 has the following successive ionization energies (in kJ/mol): IE_1 = 578, IE_2 = 1,817, IE_3 = 2,745, IE_4 = 11,577, IE_5 = 14,842. Identify Element X and write the electron configuration of its most common stable ion.

- **Answer**: The massive jump occurs between IE_3 (2,745 kJ/mol) and IE_4 (11,577 kJ/mol), showing that the 4th electron is removed from an inner core shell (n = 2) after 3 valence electrons are removed. In Period 3, the

element with 3 valence electrons is **Aluminum (Al)**. Its stable ion is Al^{3+} with electron configuration **$1s^2$ $2s^2$ $2p^6$**.